Preliminary Analysis

Data Cleaning and Exploration

```
library(readxl)
df <- read_excel("Data/Realistic_Fake_Data_Preschool_Expulsion.xlsm")
head(df)
```

```
# A tibble: 6 x 18
     ID Group Match_ID Sex   Race  Age_Months Income CASL_Exp CASL_Rec CASL_Prag
  <dbl> <chr>    <dbl> <chr> <chr>      <dbl>  <dbl>    <dbl>    <dbl>     <dbl>
1   101 At R~        1 M     Blac~         42  19000       83       74        74
2   201 Not ~        1 M     Blac~         40  24000       80       78       102
3   102 At R~        2 F     Blac~         38  28000       92       85        80
4   202 Not ~        2 F     Blac~         41  35000       90       88        85
5   103 At R~        3 M     Lati~         52  22000       92       96        72
6   203 Not ~        3 M     Lati~         49  29000       85       89        95
# i 8 more variables: CASL_SentExp <dbl>, CASL_GLAI <dbl>, Vineland_Exp <dbl>,
#   Vineland_Rec <dbl>, BRIEF_P_T <dbl>, HTKS_Score <dbl>, SDQ_Total <dbl>,
#   PERM_Score <dbl>
```

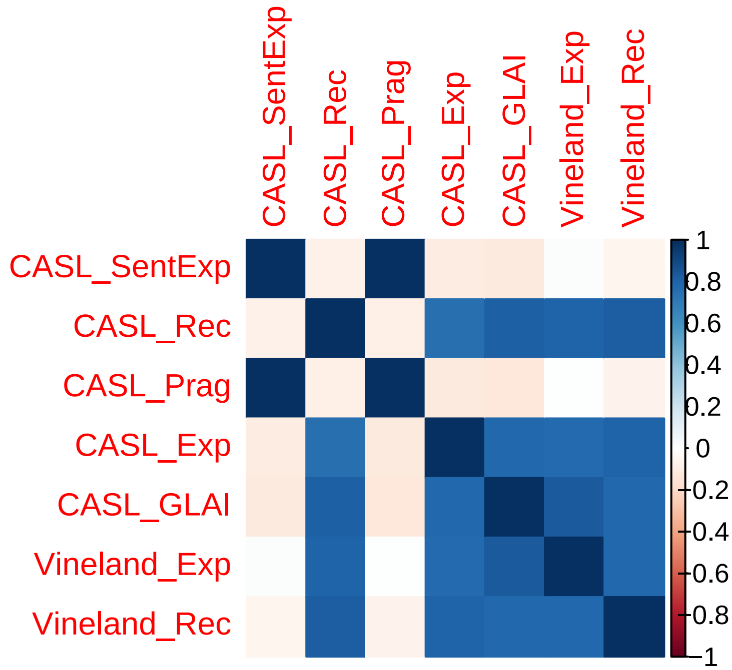
```
language <- c("CASL_SentExp", "CASL_Rec", "CASL_Prag", "CASL_Exp", "CASL_GLAI", "Vineland_Exp
behavior <- c("SDQ_Total")
cognitive_communication <- c("BRIEF_P_T", "HTKS_Score")
expulsion <- c("PERM_Score")
#colnames(df)
cor_data <- cor(df[, c(language, behavior,
                                cognitive_communication,
                                expulsion)])
cor_df <- data.frame(cor(df[, c(language, behavior,
                                cognitive_communication,
                                expulsion)]))
cor_df
```

```
             CASL_SentExp    CASL_Rec    CASL_Prag    CASL_Exp   CASL_GLAI
CASL_SentExp   1.00000000 -0.07103702  0.998426187 -0.10857774 -0.11819435
CASL_Rec      -0.07103702  1.00000000 -0.084091942  0.75564568  0.81895392
CASL_Prag      0.99842619 -0.08409194  1.000000000 -0.11780207 -0.12437002
CASL_Exp      -0.10857774  0.75564568 -0.117802072  1.00000000  0.78476537
CASL_GLAI     -0.11819435  0.81895392 -0.124370024  0.78476537  1.00000000
Vineland_Exp   0.01679093  0.80178749  0.005616079  0.77963697  0.83135845
Vineland_Rec  -0.05361046  0.82125268 -0.066524183  0.80957058  0.78102688
SDQ_Total      0.27710206  0.27699145  0.276584793  0.12124923  0.17676188
BRIEF_P_T      0.23456583  0.23556749  0.235453082  0.12258895  0.12288961
HTKS_Score     0.06708402  0.09708249  0.063084252  0.05103241  0.17332002
PERM_Score     0.31767692  0.14610887  0.318535160  0.00499057  0.05027753
             Vineland_Exp Vineland_Rec SDQ_Total BRIEF_P_T HTKS_Score
CASL_SentExp  0.016790927  -0.05361046 0.2771021 0.2345658 0.06708402
CASL_Rec      0.801787490   0.82125268 0.2769915 0.2355675 0.09708249
CASL_Prag     0.005616079  -0.06652418 0.2765848 0.2354531 0.06308425
CASL_Exp      0.779636967   0.80957058 0.1212492 0.1225890 0.05103241
CASL_GLAI     0.831358451   0.78102688 0.1767619 0.1228896 0.17332002
Vineland_Exp  1.000000000   0.78666630 0.2772783 0.2640718 0.16930724
Vineland_Rec  0.786666303   1.00000000 0.2953372 0.2412000 0.17168297
SDQ_Total     0.277278325   0.29533723 1.0000000 0.8232054 0.20633521
BRIEF_P_T     0.264071836   0.24119998 0.8232054 1.0000000 0.16683137
HTKS_Score    0.169307236   0.17168297 0.2063352 0.1668314 1.00000000
PERM_Score    0.172636174   0.16013931 0.7894974 0.8556484 0.10188995
             PERM_Score
CASL_SentExp 0.31767692
CASL_Rec     0.14610887
CASL_Prag    0.31853516
CASL_Exp     0.00499057
CASL_GLAI    0.05027753
Vineland_Exp 0.17263617
Vineland_Rec 0.16013931
SDQ_Total    0.78949739
BRIEF_P_T    0.85564842
HTKS_Score   0.10188995
PERM_Score   1.00000000
```

```
library(corrplot)
```

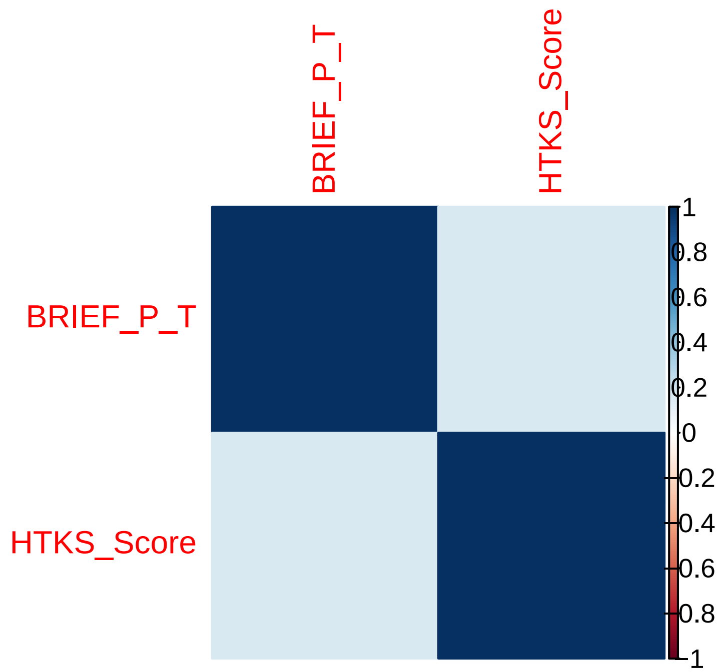
```
corrplot 0.95 loaded
```

```r
corrplot(cor_data[language, language],
         method = "color")
```



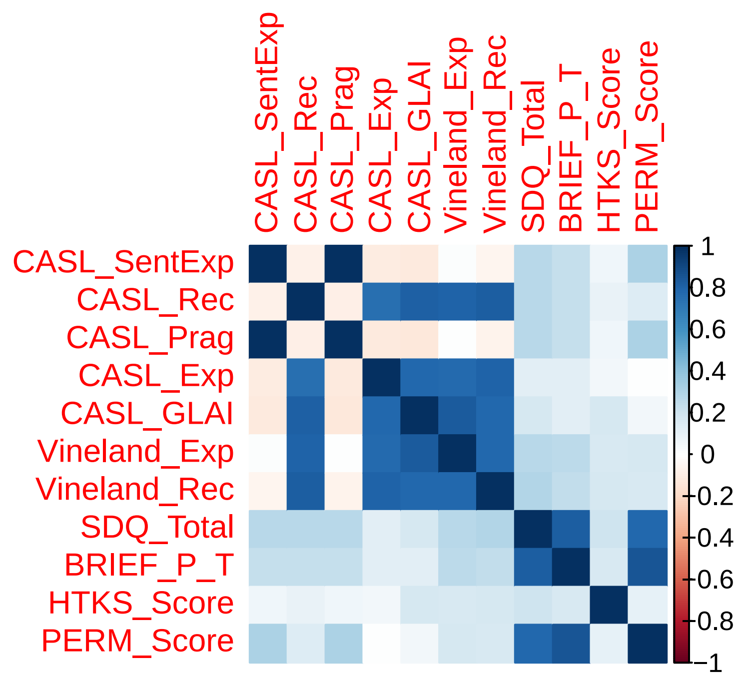
Correlation (bi-variate) of all variables for language is almost 1

```r
corrplot(cor_data[cognitive_communication, cognitive_communication],
         method = "color")
```



Correlation of all our (potential) covariates:

```
corrplot(cor_data, method = "color")
```



```r
cor_df
```

```
             CASL_SentExp    CASL_Rec    CASL_Prag    CASL_Exp   CASL_GLAI
CASL_SentExp   1.00000000 -0.07103702  0.998426187 -0.10857774 -0.11819435
CASL_Rec      -0.07103702  1.00000000 -0.084091942  0.75564568  0.81895392
CASL_Prag      0.99842619 -0.08409194  1.000000000 -0.11780207 -0.12437002
CASL_Exp      -0.10857774  0.75564568 -0.117802072  1.00000000  0.78476537
CASL_GLAI     -0.11819435  0.81895392 -0.124370024  0.78476537  1.00000000
Vineland_Exp   0.01679093  0.80178749  0.005616079  0.77963697  0.83135845
Vineland_Rec  -0.05361046  0.82125268 -0.066524183  0.80957058  0.78102688
SDQ_Total      0.27710206  0.27699145  0.276584793  0.12124923  0.17676188
BRIEF_P_T      0.23456583  0.23556749  0.235453082  0.12258895  0.12288961
HTKS_Score     0.06708402  0.09708249  0.063084252  0.05103241  0.17332002
PERM_Score     0.31767692  0.14610887  0.318535160  0.00499057  0.05027753
             Vineland_Exp Vineland_Rec SDQ_Total BRIEF_P_T HTKS_Score
CASL_SentExp  0.016790927  -0.05361046 0.2771021 0.2345658 0.06708402
CASL_Rec      0.801787490   0.82125268 0.2769915 0.2355675 0.09708249
CASL_Prag     0.005616079  -0.06652418 0.2765848 0.2354531 0.06308425
CASL_Exp      0.779636967   0.80957058 0.1212492 0.1225890 0.05103241
CASL_GLAI     0.831358451   0.78102688 0.1767619 0.1228896 0.17332002
Vineland_Exp  1.000000000   0.78666630 0.2772783 0.2640718 0.16930724
Vineland_Rec  0.786666303   1.00000000 0.2953372 0.2412000 0.17168297
SDQ_Total     0.277278325   0.29533723 1.0000000 0.8232054 0.20633521
BRIEF_P_T     0.264071836   0.24119998 0.8232054 1.0000000 0.16683137
HTKS_Score    0.169307236   0.17168297 0.2063352 0.1668314 1.00000000
PERM_Score    0.172636174   0.16013931 0.7894974 0.8556484 0.10188995
             PERM_Score
CASL_SentExp 0.31767692
CASL_Rec     0.14610887
CASL_Prag    0.31853516
CASL_Exp     0.00499057
CASL_GLAI    0.05027753
Vineland_Exp 0.17263617
Vineland_Rec 0.16013931
SDQ_Total    0.78949739
BRIEF_P_T    0.85564842
HTKS_Score   0.10188995
PERM_Score   1.00000000
```

```r
summary(df[, language])
```

```
  CASL_SentExp        CASL_Rec          CASL_Prag          CASL_Exp
```

```
 Min.   : 67.00   Min.   : 71.00   Min.   : 66.00   Min.   : 64.00  
 1st Qu.: 78.75   1st Qu.: 84.75   1st Qu.: 77.75   1st Qu.: 81.75  
 Median : 84.50   Median : 90.00   Median : 83.50   Median : 87.00  
 Mean   : 86.74   Mean   : 89.81   Mean   : 85.81   Mean   : 87.04  
 3rd Qu.: 94.00   3rd Qu.: 96.25   3rd Qu.: 93.25   3rd Qu.: 92.25  
 Max.   :116.00   Max.   :106.00   Max.   :115.00   Max.   :108.00  
   CASL_GLAI      Vineland_Exp     Vineland_Rec   
 Min.   : 67.0   Min.   : 55.00   Min.   : 69.00  
 1st Qu.: 82.0   1st Qu.: 82.00   1st Qu.: 83.00  
 Median : 88.0   Median : 87.50   Median : 89.50  
 Mean   : 87.1   Mean   : 87.35   Mean   : 89.22  
 3rd Qu.: 92.0   3rd Qu.: 92.00   3rd Qu.: 96.00  
 Max.   :103.0   Max.   :108.00   Max.   :110.00  
```

```r
#Language Variables
library(dplyr)
```

```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```r
df %>% group_by(Group) %>% summarise(
  Median_CASL_SentExp = median(CASL_SentExp),
  Median_CASL_Exp = median(CASL_Exp),
  Median_CASL_Rec = median(CASL_Rec),
  Median_CASL_Prag = median(CASL_Prag),
  Median_CASL_GLAI = median(CASL_GLAI),
  Median_Vineland_Exp = median(Vineland_Exp),
  Median_Vineland_Rec = median(Vineland_Rec))
```

```
# A tibble: 2 x 8
  Group     Median_CASL_SentExp Median_CASL_Exp Median_CASL_Rec Median_CASL_Prag
  <chr>                   <dbl>           <dbl>           <dbl>            <dbl>
```

```
1 At Risk                     78.5             89               92              77.5
2 Not At R~                   94               87               89              93.5
# i 3 more variables: Median_CASL_GLAI <dbl>, Median_Vineland_Exp <dbl>,
#   Median_Vineland_Rec <dbl>
```

```r
#Difference in CASL_Exp and CASL_SentExp:
diff <- abs(df$CASL_Exp - df$CASL_SentExp)
mean(diff)
```

```
[1] 12.04412
```

```r
#Cognitive Communication
df %>% group_by(Group) %>% summarise(Median_Brief_P_T = median(BRIEF_P_T),
                                     Median_HTKS = median(HTKS_Score))
```

```
# A tibble: 2 x 3
  Group        Median_Brief_P_T Median_HTKS
  <chr>                   <dbl>       <dbl>
1 At Risk                    58        17.5
2 Not At Risk                59        18
```

```r
#Behavior and PERM Score
df %>% group_by(Group) %>% summarise(Median_SDQ = median(SDQ_Total),
                                     Median_PERM = median(PERM_Score))
```

```
# A tibble: 2 x 3
  Group        Median_SDQ Median_PERM
  <chr>             <dbl>       <dbl>
1 At Risk            11.5        2.56
2 Not At Risk        15.5        3.26
```

```r
df$Avg_Lang <- rowMeans(df[, language], na.rm = F)
cor(df$Avg_Lang, df$BRIEF_P_T)
```

```
[1] 0.3231168
```

For now I am keeping BRIEF_P_T and dropping HTKS_Score (both variables have a similarly strong relationship (with other covariates), my intuition is they have the same context in terms of latent factors)

```r
#Drop Variables until VIF < 20
library(car)

vars_of_interest <- c(language, behavior, cognitive_communication)

drop_until_vif_thresh <- function(data, vars, thresh){
  data[, vars] <- as.data.frame(scale(data[, vars]))
  data$Y <- ifelse(data$Group == "At Risk", 1, 0)
  data_of_interest <- data[, c(vars, "Y")]
  model <- glm(data = data_of_interest, formula = Y ~ ., family = "binomial")
  vif_score <- vif(model)
  cols_dropped <- c()
  cols_dropped_with_vif <- c()
  while(max(vif_score)>10){
    max_vif_col <- names(vif_score[vif_score == max(vif_score)])
    max_vif_value <- vif_score[vif_score == max(vif_score)]
    cols_dropped <- c(cols_dropped, max_vif_col)
    cols_dropped_with_vif <- c(cols_dropped_with_vif, max_vif_value)
    data_of_interest <- data_of_interest %>% dplyr::select(-all_of(max_vif_col))
    model <- glm(data = data_of_interest, formula = Y ~ ., family = "binomial")

    vif_score <- vif(model)
  }
  results <- list(col_to_drop = cols_dropped, vif_values = cols_dropped_with_vif, model_sele
}
```

```r
res_list <- drop_until_vif_thresh(df, vars_of_interest, 5)
```

```r
summary(res_list$model_selected)
```

```
Call:
glm(formula = Y ~ ., family = "binomial", data = data_of_interest)

Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept)   -0.5139     0.5090  -1.010  0.31259
CASL_SentExp  -4.2938     1.4077  -3.050  0.00229 **
CASL_Rec      -0.1197     0.9727  -0.123  0.90208
CASL_Exp      -0.3152     0.8451  -0.373  0.70915
CASL_GLAI      0.7121     1.1058   0.644  0.51961
```

```
Vineland_Exp   0.5138     0.8963   0.573  0.56646
SDQ_Total     -1.8058     1.0418  -1.733  0.08303 .
BRIEF_P_T      1.2555     0.9967   1.260  0.20780
HTKS_Score     0.1460     0.4161   0.351  0.72565
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 94.268  on 67  degrees of freedom
Residual deviance: 38.343  on 59  degrees of freedom
AIC: 56.343

Number of Fisher Scoring iterations: 7
```

```
#Conducting same test with avg scores for language
vars_of_interest_avg_lang <- c("Avg_Lang", behavior, cognitive_communication)
res_list_with_avg_lang <- drop_until_vif_thresh(df,
                                                vars_of_interest_avg_lang)
summary(res_list_with_avg_lang$model_selected)
```

```
Call:
glm(formula = Y ~ ., family = "binomial", data = data_of_interest)

Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept) -0.006201   0.258161  -0.024    0.981
Avg_Lang    -0.384289   0.292288  -1.315    0.189
SDQ_Total   -0.687251   0.483998  -1.420    0.156
BRIEF_P_T    0.175528   0.468784   0.374    0.708
HTKS_Score   0.182629   0.267462   0.683    0.495

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 94.268  on 67  degrees of freedom
Residual deviance: 85.983  on 63  degrees of freedom
AIC: 95.983

Number of Fisher Scoring iterations: 4
```

Based on research and analysis, I think exploratory factor analysis should be done.

PERM: Quantify risk of pre-school expulsion

Raw Variables

Goal 1: Analyze differences in At-Risk and Not at-Risk groups for language and cognitive communication variables

Approach 1: MANOVA with the continuous variables as predictors

```
df$Cognitive_Avg <- (df$BRIEF_P_T + df$HTKS_Score)/2
head(df)
```

```
# A tibble: 6 x 20
     ID Group Match_ID Sex   Race  Age_Months Income CASL_Exp CASL_Rec CASL_Prag
  <dbl> <chr>    <dbl> <chr> <chr>      <dbl>  <dbl>    <dbl>    <dbl>     <dbl>
1   101 At R~        1 M     Blac~         42  19000       83       74        74
2   201 Not ~        1 M     Blac~         40  24000       80       78       102
3   102 At R~        2 F     Blac~         38  28000       92       85        80
4   202 Not ~        2 F     Blac~         41  35000       90       88        85
5   103 At R~        3 M     Lati~         52  22000       92       96        72
6   203 Not ~        3 M     Lati~         49  29000       85       89        95
# i 10 more variables: CASL_SentExp <dbl>, CASL_GLAI <dbl>, Vineland_Exp <dbl>,
#   Vineland_Rec <dbl>, BRIEF_P_T <dbl>, HTKS_Score <dbl>, SDQ_Total <dbl>,
#   PERM_Score <dbl>, Avg_Lang <dbl>, Cognitive_Avg <dbl>
```

```
y1 <- df$CASL_GLAI#Language
y2 <- df$Cognitive_Avg#Cognitive Communication
y3 <- df$SDQ_Total#Behavior
x <- ifelse(df$Group == "At Risk", 1, 0)
model <- manova(cbind(y1, y2, y3) ~ x)

summary(model, test = "Pillai")
```

```
          Df  Pillai approx F num Df den Df  Pr(>F)
x          1 0.15828   4.0116      3     64 0.01114 *
Residuals 66
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
summary.aov(model)
```

```
 Response y1 :
            Df Sum Sq Mean Sq F value  Pr(>F)
x            1  174.7 174.721  3.0254 0.08663 .
Residuals   66 3811.6  57.751
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

 Response y2 :
            Df  Sum Sq Mean Sq F value Pr(>F)
x            1   30.44  30.445  1.3416 0.2509
Residuals   66 1497.71  22.693

 Response y3 :
            Df  Sum Sq Mean Sq F value  Pr(>F)
x            1  187.78 187.779  6.1615 0.01561 *
Residuals   66 2011.44  30.476
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Bonferroni gives alpha = 0.0167 (0.05/3), none of these are significant. These are just individual T-tests

Linear Discriminant Analysis to find which covariate is driving the phenomenon above

```r
library(MASS)
```

```
Attaching package: 'MASS'


The following object is masked from 'package:dplyr':

    select
```

```r
lda <- lda(x ~ scale(y1) + scale(y2) + scale(y3))

print(lda)
```

```
Call:
lda(x ~ scale(y1) + scale(y2) + scale(y3))

Prior probabilities of groups:
  0   1
0.5 0.5

Group means:
   scale(y1)  scale(y2)  scale(y3)
0 -0.2078122  0.1401057  0.2900498
1  0.2078122 -0.1401057 -0.2900498

Coefficients of linear discriminants:
                 LD1
scale(y1)  0.7182663
scale(y2)  0.1885934
scale(y3) -1.0493814
```

Behavior is driving this.

Approach 2: Y = {At Risk: 1, Not At Risk: 0}, X = Continuous Variables for language and cognitive communication variable. Fit a logistic regression with Y~X to find associations. Based on how the experiment is designed we can think about causal effects (not sure about causation).

Goal 2: Nested Models: Group ~ Language v/s Group ~ Language + Behavior

Add cognitive communication here - should be correlated with behavior

CASL: Comprehensive Assessment of Spoken Language - GLAI is General Language Ability Index and seems like a composite score so I'll use that in the model to capture the effect of language.

SentExp: Sentence Expression sub-test of CASL

Exp: Expressiveness: Different kind of expressiveness from SentExp

Prag: Pragmatic

Rec: Receptive - understanding language

Vineland is another standardized test for language abilities

PERM: represents the Personal subdomain of adaptive behavior, which contributes to overall adaptive behavior scores. - just rated by the teacher

Boys and kids from lower income should have lower cognitive scores

CASL and Vineland are normally distributed

SDQ_Total : Behavior

-All of this is stuff I found online, I need to get this validated.

Model just with Language (reduced model):

$p_i = Probabibility\ of\ i^{th}\ child\ being\ at\ risk$

$X_i = CASL\ GLAI\ score\ for\ the\ i^{th}\ child$

Model:

$$log\frac{p_i}{1-p_i} = \beta_0 + \beta_1 * X_i$$

i = {1,2,……, 68}

```
#Reduced Model
cols_of_interest <- c("CASL_GLAI", "SDQ_Total")
df_nested_models <- df[, c("Group", cols_of_interest)]
df_nested_models[, cols_of_interest] <- scale(df[, cols_of_interest])
df_nested_models$Y <- ifelse(df_nested_models$Group == "At Risk", 1, 0)
df$Y <- ifelse(df$Group == "At Risk", 1, 0)
m1 <- glm(data = df_nested_models, formula = Y ~ CASL_GLAI, family = "binomial")
summary(m1)
```

```
Call:
glm(formula = Y ~ CASL_GLAI, family = "binomial", data = df_nested_models)

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept) -0.0009311  0.2480417  -0.004    0.997
CASL_GLAI    0.4405648  0.2606910   1.690    0.091 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 94.268  on 67  degrees of freedom
Residual deviance: 91.223  on 66  degrees of freedom
AIC: 95.223
```

```
Number of Fisher Scoring iterations: 4
```

A unit increase in GLAI score correlates (don't know about design) with a reduction of odds of child being at risk, there's a ~% ((e^beta_1 - 1)*100) decrease in odds with every unit increase in GLAI (after scaling).

-the coefficient should be negative

```
m1_unscaled <- glm(data = df, formula = Y ~ SDQ_Total, family = "binomial")
summary(m1_unscaled)
```

```
Call:
glm(formula = Y ~ SDQ_Total, family = "binomial", data = df)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  1.61527    0.73455   2.199   0.0279 *
SDQ_Total   -0.11068    0.04763  -2.324   0.0201 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 94.268  on 67  degrees of freedom
Residual deviance: 88.246  on 66  degrees of freedom
AIC: 92.246

Number of Fisher Scoring iterations: 4
```

If GLAI is unscaled then ((e^{β_1} - 1)*100) % decrease in odds with every unit increase in GLAI.

Main Effects model with Language + Behavior:

$p_i = Probability\ of\ i^{th}\ child\ being\ at\ risk$

$X_i = CASL\ GLAI\ score\ for\ the\ i^{th}\ child$

$Z_i = SDQ\ Total\ score\ for\ the\ ith\ child$

Model:

$$log \frac{p_i}{1-p_i} = \beta_0 \ + \beta_1 * X_i \ + \beta_2 * Z_i$$

i = {1,2,……, 68}

```r
#Full Model - with main effect
m2 <- glm(data = df_nested_models, formula = Y ~ CASL_GLAI + SDQ_Total, family = "binomial")
summary(m2)
```

```
Call:
glm(formula = Y ~ CASL_GLAI + SDQ_Total, family = "binomial",
    data = df_nested_models)

Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept) -0.007689   0.264389  -0.029  0.97680
CASL_GLAI    0.643946   0.291610   2.208  0.02723 *
SDQ_Total   -0.816246   0.307480  -2.655  0.00794 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 94.268  on 67  degrees of freedom
Residual deviance: 82.749  on 65  degrees of freedom
AIC: 88.749

Number of Fisher Scoring iterations: 3
```

Lower GLAI ~ Higher risk

High SDQ ~ Worse Behavior - higher risk

```r
lr_covariates <- lm(data = df_nested_models, formula = SDQ_Total ~ CASL_GLAI)
MSE <- sum(resid(lr_covariates) ^ 2)/(68-2)
summary(lr_covariates)
```

```
Call:
lm(formula = SDQ_Total ~ CASL_GLAI, data = df_nested_models)

Residuals:
    Min      1Q  Median      3Q     Max
-2.1107 -0.5905 -0.1846  0.7499  2.3992
```

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) 1.586e-16  1.203e-01   0.000    1.000
CASL_GLAI   1.768e-01  1.212e-01   1.459    0.149

Residual standard error: 0.9917 on 66 degrees of freedom
Multiple R-squared:  0.03124,   Adjusted R-squared:  0.01657
F-statistic: 2.129 on 1 and 66 DF,  p-value: 0.1493
```

```
#LRT for nested models
anova(m1, m2, test = "LRT")
```

```
Analysis of Deviance Table

Model 1: Y ~ CASL_GLAI
Model 2: Y ~ CASL_GLAI + SDQ_Total
  Resid. Df Resid. Dev Df Deviance Pr(>Chi)
1        66     91.223
2        65     82.749  1   8.4735 0.003604 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

- According to LRT, adding Behavior explains significantly additional variability in log(odds)
- Wald's test agrees with this inference

Nested Model:

Interaction Effects model with Language and Behavior:

$p_i = Probability\ of\ i^{th}\ child\ being\ at\ risk$

$X_i = CASL\ GLAI\ score\ for\ the\ i^{th}\ child$

$Z_i = SDQ\ Total\ score\ for\ the\ ith\ child$

Model:

$$log\frac{p_i}{1-p_i} = \beta_0\ + \beta_1 * X_i\ + \beta_2 * Z_i\ + \beta_3 * X_i * Z_i$$

i = {1,2,……, 68}

```r
m3 <- glm(data = df_nested_models, formula = Y ~ CASL_GLAI*SDQ_Total, family = "binomial")
summary(m3)
```

```
Call:
glm(formula = Y ~ CASL_GLAI * SDQ_Total, family = "binomial",
    data = df_nested_models)

Coefficients:
                     Estimate Std. Error z value Pr(>|z|)
(Intercept)         -0.004183   0.281874  -0.015  0.98816
CASL_GLAI            0.642732   0.293518   2.190  0.02854 *
SDQ_Total           -0.815286   0.308458  -2.643  0.00822 **
CASL_GLAI:SDQ_Total -0.011816   0.329764  -0.036  0.97142
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 94.268  on 67  degrees of freedom
Residual deviance: 82.748  on 64  degrees of freedom
AIC: 90.748

Number of Fisher Scoring iterations: 3
```

Compare m3 v/s m1:

```r
anova(m1, m3, test = "LRT")
```

```
Analysis of Deviance Table

Model 1: Y ~ CASL_GLAI
Model 2: Y ~ CASL_GLAI * SDQ_Total
  Resid. Df Resid. Dev Df Deviance Pr(>Chi)
1        66     91.223
2        64     82.748  2   8.4748  0.01445 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Adding SDQ_Total (with interaction) significantly improves when compared to model with just GLAI

Compare m2 v/s m3:

```r
anova(m2, m3, test = "LRT")
```

```
Analysis of Deviance Table

Model 1: Y ~ CASL_GLAI + SDQ_Total
Model 2: Y ~ CASL_GLAI * SDQ_Total
  Resid. Df Resid. Dev Df  Deviance Pr(>Chi)
1        65     82.749                      
2        64     82.748  1 0.0012831   0.9714
```

Adding interaction is not a significant improvement over just main effects.

Factor Analysis

Language

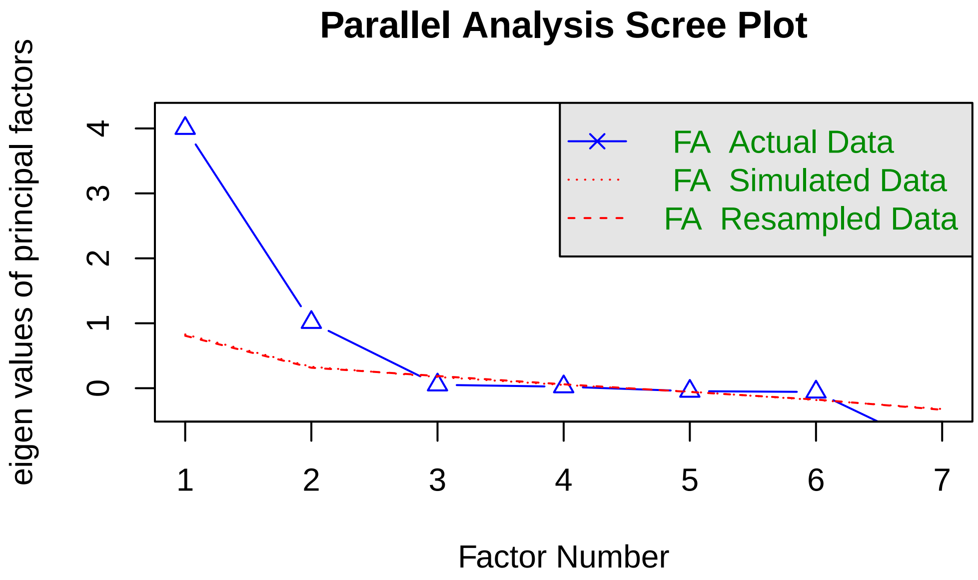
```r
library(psych)
```

```
Attaching package: 'psych'


The following object is masked from 'package:car':

    logit
```

```r
parallel_model <- fa.parallel(df[, language],
                              fm = "pa",
                              fa = "fa",
                              n.iter = 50,
                              main = "Parallel Analysis Scree Plot")
```

```
Warning in fac(r = r, nfactors = nfactors, n.obs = n.obs, rotate = rotate, : An
ultra-Heywood case was detected.  Examine the results carefully
Warning in fac(r = r, nfactors = nfactors, n.obs = n.obs, rotate = rotate, : An
ultra-Heywood case was detected.  Examine the results carefully
Warning in fac(r = r, nfactors = nfactors, n.obs = n.obs, rotate = rotate, : An
ultra-Heywood case was detected.  Examine the results carefully
```

```
Warning in fac(r = r, nfactors = nfactors, n.obs = n.obs, rotate = rotate, : An
ultra-Heywood case was detected.  Examine the results carefully
Warning in fac(r = r, nfactors = nfactors, n.obs = n.obs, rotate = rotate, : An
ultra-Heywood case was detected.  Examine the results carefully
Warning in fac(r = r, nfactors = nfactors, n.obs = n.obs, rotate = rotate, : An
ultra-Heywood case was detected.  Examine the results carefully
Warning in fac(r = r, nfactors = nfactors, n.obs = n.obs, rotate = rotate, : An
ultra-Heywood case was detected.  Examine the results carefully
Warning in fac(r = r, nfactors = nfactors, n.obs = n.obs, rotate = rotate, : An
ultra-Heywood case was detected.  Examine the results carefully
Warning in fac(r = r, nfactors = nfactors, n.obs = n.obs, rotate = rotate, : An
ultra-Heywood case was detected.  Examine the results carefully
Warning in fac(r = r, nfactors = nfactors, n.obs = n.obs, rotate = rotate, : An
ultra-Heywood case was detected.  Examine the results carefully
Warning in fac(r = r, nfactors = nfactors, n.obs = n.obs, rotate = rotate, : An
ultra-Heywood case was detected.  Examine the results carefully
```




```
Parallel analysis suggests that the number of factors =  2  and the number of components =
```

we need 2 factors

```r
efa_model_lang <- fa(r = df[, language],
                     nfactors = 2,
                     fm = "pa",
                     rotate = "oblimin",
                     scores = "regression")
```

```
Loading required namespace: GPArotation
```

```r
print(efa_model_lang)
```

```
Factor Analysis using method =  pa
Call: fa(r = df[, language], nfactors = 2, rotate = "oblimin", scores = "regression",
    fm = "pa")
Standardized loadings (pattern matrix) based upon correlation matrix
               PA1   PA2   h2      u2 com
CASL_SentExp  0.01  1.00 1.00 0.00065   1
CASL_Rec      0.90 -0.01 0.80 0.19706   1
CASL_Prag    -0.01  1.00 1.00 0.00218   1
CASL_Exp      0.87 -0.04 0.76 0.23916   1
CASL_GLAI     0.90 -0.05 0.82 0.18116   1
Vineland_Exp  0.91  0.08 0.82 0.18483   1
Vineland_Rec  0.90  0.01 0.80 0.19769   1

                       PA1  PA2
SS loadings           3.99 2.01
Proportion Var        0.57 0.29
Cumulative Var        0.57 0.86
Proportion Explained  0.67 0.33
Cumulative Proportion 0.67 1.00

 With factor correlations of
      PA1   PA2
PA1  1.00 -0.08
PA2 -0.08  1.00

Mean item complexity =  1
Test of the hypothesis that 2 factors are sufficient.

df null model =  21  with the objective function =  10.98 with Chi Square =  701.13
df of  the model are 8  and the objective function was  0.15
```

```
The root mean square of the residuals (RMSR) is  0.01
The df corrected root mean square of the residuals is  0.02

The harmonic n.obs is  68 with the empirical chi square  0.25  with prob <  1
The total n.obs was  68  with Likelihood Chi Square =  9.56  with prob <  0.3

Tucker Lewis Index of factoring reliability =  0.994
RMSEA index =  0.051  and the 90 % confidence intervals are  0 0.159
BIC =  -24.2
Fit based upon off diagonal values = 1
Measures of factor score adequacy
                                                   PA1 PA2
Correlation of (regression) scores with factors   0.98   1
Multiple R square of scores with factors          0.95   1
Minimum correlation of possible factor scores     0.91   1
```

```
print(efa_model_lang$loadings)
```

```
Loadings:
             PA1    PA2
CASL_SentExp         1.000
CASL_Rec      0.896
CASL_Prag            0.998
CASL_Exp      0.868
CASL_GLAI     0.900
Vineland_Exp  0.906
Vineland_Rec  0.896

                 PA1   PA2
SS loadings    3.988 2.008
Proportion Var 0.570 0.287
Cumulative Var 0.570 0.857
```

We can observe that CASL_SentExp and CASL_Prag load very strongly on the second latent factor, while everything else loads very strongly on the first latent factor. I will be using these latent factors for achieving the objectives.

```
#W_i * X_i = latent_factor_i
print(efa_model_lang$weights)
```

```
                    PA1          PA2
CASL_SentExp  0.2047457  0.780902092
CASL_Rec      0.2049206 -0.002200388
CASL_Prag    -0.2091872  0.220249775
CASL_Exp      0.1701574  0.002911683
CASL_GLAI     0.2379724  0.010765002
Vineland_Exp  0.2276132 -0.006855978
Vineland_Rec  0.2217542 -0.007511403
```

```r
df$Language_LatentFactor_1 <- efa_model_lang$scores[, 1]
df$Language_LatentFactor_2 <- efa_model_lang$scores[, 2]
head(df)
```

```
# A tibble: 6 x 23
     ID Group Match_ID Sex   Race  Age_Months Income CASL_Exp CASL_Rec CASL_Prag
  <dbl> <chr>    <dbl> <chr> <chr>      <dbl>  <dbl>    <dbl>    <dbl>     <dbl>
1   101 At R~        1 M     Blac~         42  19000       83       74        74
2   201 Not ~        1 M     Blac~         40  24000       80       78       102
3   102 At R~        2 F     Blac~         38  28000       92       85        80
4   202 Not ~        2 F     Blac~         41  35000       90       88        85
5   103 At R~        3 M     Lati~         52  22000       92       96        72
6   203 Not ~        3 M     Lati~         49  29000       85       89        95
# i 13 more variables: CASL_SentExp <dbl>, CASL_GLAI <dbl>, Vineland_Exp <dbl>,
#   Vineland_Rec <dbl>, BRIEF_P_T <dbl>, HTKS_Score <dbl>, SDQ_Total <dbl>,
#   PERM_Score <dbl>, Avg_Lang <dbl>, Cognitive_Avg <dbl>, Y <dbl>,
#   Language_LatentFactor_1 <dbl>, Language_LatentFactor_2 <dbl>
```

Cognitive Communication

```r
efa_model_cog_comm <- fa(r = df[, cognitive_communication],
                     nfactors = 1,
                     fm = "pa",
                     rotate = "oblimin",
                     scores = "regression")

print(efa_model_cog_comm)
```

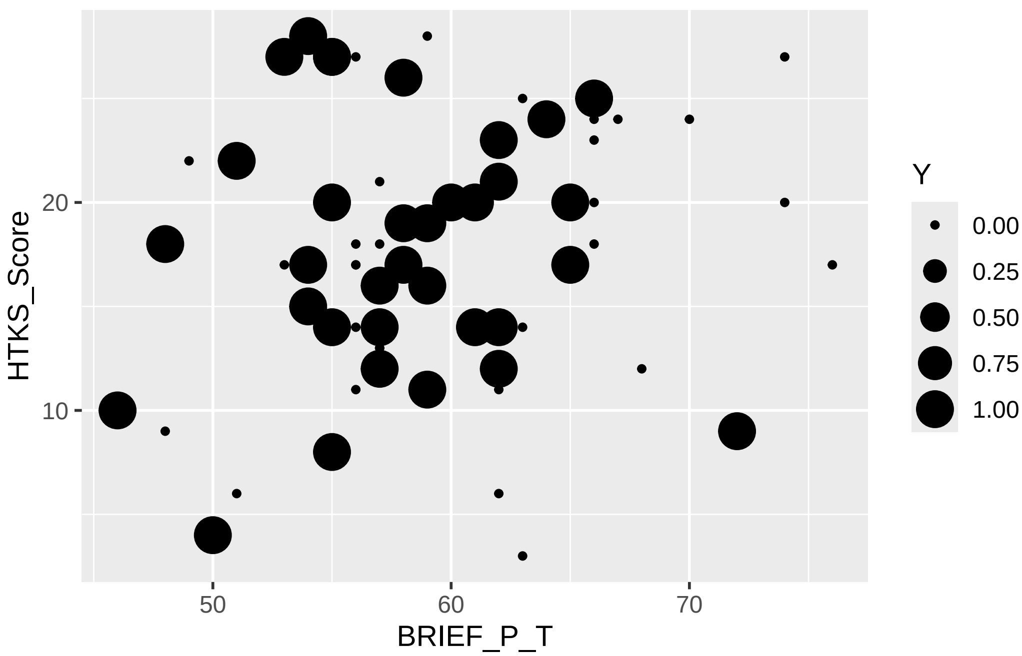
```
Factor Analysis using method =  pa
Call: fa(r = df[, cognitive_communication], nfactors = 1, rotate = "oblimin",
    scores = "regression", fm = "pa")
```

```
Standardized loadings (pattern matrix) based upon correlation matrix
            PA1   h2   u2 com
BRIEF_P_T  0.41 0.17 0.83   1
HTKS_Score 0.41 0.17 0.83   1

                PA1
SS loadings    0.33
Proportion Var 0.17

Mean item complexity =  1
Test of the hypothesis that 1 factor is sufficient.

df null model =  1  with the objective function =  0.03 with Chi Square =  1.85
df of  the model are -1  and the objective function was  0

The root mean square of the residuals (RMSR) is  0
The df corrected root mean square of the residuals is  NA

The harmonic n.obs is  68 with the empirical chi square  0  with prob <  NA
The total n.obs was  68  with Likelihood Chi Square =  0  with prob <  NA

Tucker Lewis Index of factoring reliability =  2.205
Fit based upon off diagonal values = 1
Measures of factor score adequacy
                                                   PA1
Correlation of (regression) scores with factors   0.53
Multiple R square of scores with factors          0.29
Minimum correlation of possible factor scores    -0.43
```

```r
library(ggplot2)
```

```

Attaching package: 'ggplot2'


The following objects are masked from 'package:psych':

    %+%, alpha
```

```r
ggplot(df, aes(x = BRIEF_P_T, y = HTKS_Score)) +
  geom_point(aes(size = Y))
```



Both variables load moderately strongly on the latent factor, this graph was just to check whether they were too correlated since their loadings are almost equal, it's nothing to worry about.

```
#W_i * X_i = latent_factor_i
print(efa_model_cog_comm$weights)
```

```
                 PA1
BRIEF_P_T  0.3497656
HTKS_Score 0.3497656
```

```
df$Cognitive_Communication_LatentFactor <- efa_model_cog_comm$scores[, 1]
```

The independent variables according to sub-categories are as follows:

1. Language: Language_Latent_Factor_1, Language_Latent_Factor_2
2. Cognitive Communication: Cognitive_Communication_Latent_Factor
3. Behavior: SDQ_Total
4. Expulsion Risk Based on Teacher (Qualitative score?): PERM_Score

The dependent variable is Y and it is encoded as follows:

1. Y = 1 if Group = "At Risk"

2. Y = 0 if Group = "Not At Risk"

The 2 objectives are:

1. Analyze differences in At-Risk and Not at-Risk groups for language and cognitive communication variables
2. Nested Models: Group ~ Language v/s Group ~ Language + Behavior

Goal 1 (Factor Analysis)

Difference in groups (MANOVA):

```r
y1 <- df$Language_LatentFactor_1#Language
y2 <- df$Language_LatentFactor_2#Language

y3 <- df$Cognitive_Communication_LatentFactor#Cognitive Communication
y4 <- df$SDQ_Total#Behavior
y5 <- df$PERM_Score

Y <- df$Y

model <- manova(cbind(y1, y2, y3, y4, y5) ~ Y)

summary(model, test = "Pillai")
```

```
          Df  Pillai approx F num Df den Df    Pr(>F)
Y          1 0.55133   15.237      5     62 9.497e-10 ***
Residuals 66
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
summary.aov(model)
```

```
 Response y1 :
            Df Sum Sq Mean Sq F value Pr(>F)
Y            1  2.177 2.17675  2.3296 0.1317
Residuals   66 61.670 0.93439

 Response y2 :
            Df Sum Sq Mean Sq F value    Pr(>F)
Y            1 33.968  33.968   67.93 9.771e-12 ***
```

```
Residuals   66 33.003   0.500
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

 Response y3 :
            Df  Sum Sq Mean Sq F value Pr(>F)
Y            1  0.3733 0.37328  1.3136 0.2559
Residuals   66 18.7546 0.28416

 Response y4 :
            Df  Sum Sq Mean Sq F value  Pr(>F)
Y            1  187.78 187.779  6.1615 0.01561 *
Residuals   66 2011.44  30.476
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

 Response y5 :
            Df Sum Sq Mean Sq F value   Pr(>F)
Y            1  6.024  6.0244  7.9938 0.006207 **
Residuals   66 49.740  0.7536
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

LDA:

```r
lda <- lda(x ~ scale(y1) + scale(y2) + scale(y3) + scale(y4) + scale(y5))

print(lda)
```

```
Call:
lda(x ~ scale(y1) + scale(y2) + scale(y3) + scale(y4) + scale(y5))

Prior probabilities of groups:
  0   1
0.5 0.5

Group means:
   scale(y1)  scale(y2)  scale(y3)  scale(y4)  scale(y5)
0 -0.1832814  0.7069264  0.1386645  0.2900498  0.3262588
1  0.1832814 -0.7069264 -0.1386645 -0.2900498 -0.3262588

Coefficients of linear discriminants:
```

```
                  LD1
scale(y1)  0.3122863
scale(y2) -1.2978354
scale(y3)  0.2221846
scale(y4) -0.3237177
scale(y5) -0.1663953
```

Goal 2 (Factor Analysis)

```
m1 <- glm(data = df, formula = Y ~ Language_LatentFactor_1 + Language_LatentFactor_2 + Cogni
summary(m1)
```

```
Call:
glm(formula = Y ~ Language_LatentFactor_1 + Language_LatentFactor_2 +
    Cognitive_Communication_LatentFactor + SDQ_Total + PERM_Score,
    family = "binomial", data = df)

Coefficients:
                                     Estimate Std. Error z value Pr(>|z|)
(Intercept)                            2.3980     1.9234   1.247 0.212475
Language_LatentFactor_1                0.6647     0.4645   1.431 0.152454
Language_LatentFactor_2               -3.6262     1.0701  -3.389 0.000702 ***
Cognitive_Communication_LatentFactor   1.1566     1.0112   1.144 0.252683
SDQ_Total                             -0.1618     0.1450  -1.116 0.264627
PERM_Score                            -0.1189     0.7463  -0.159 0.873441
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 94.268  on 67  degrees of freedom
Residual deviance: 40.265  on 62  degrees of freedom
AIC: 52.265

Number of Fisher Scoring iterations: 7
```

```
m2 <- glm(data = df, formula = Y ~ Language_LatentFactor_1 + Language_LatentFactor_2 , family
m3 <- glm(data = df, formula = Y ~ Language_LatentFactor_1 + Language_LatentFactor_2 + SDQ_To

anova(m2, m3, test = "Chisq")
```

```
Analysis of Deviance Table

Model 1: Y ~ Language_LatentFactor_1 + Language_LatentFactor_2
Model 2: Y ~ Language_LatentFactor_1 + Language_LatentFactor_2 + SDQ_Total
  Resid. Df Resid. Dev Df Deviance Pr(>Chi)
1        65     43.010
2        64     41.593  1   1.4175   0.2338
```

Adding behavior doesn't explain significantly additional variability.